RuFaS Data Collection App User Guide

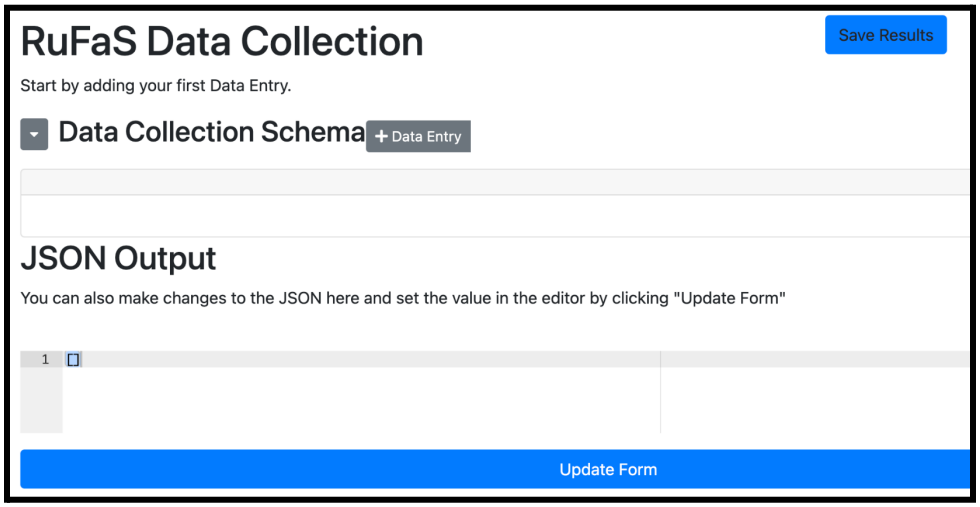
The RuFaS Data Collection App allows users to easily collect all necessary farm data in one location to be able to run full-farm simulations using the RuFaS model. This app is cross-platform, meaning users can run it from any OS such as Windows or Mac. It is fully functional with or without an internet connection - allowing users to collect data from any location.

There are 10 sections of the farm on which this app will help you collect data, each with its own Data Entry form (or schema) available from the main page of the app.

- Animals
- Feeds
- Manure Storage and Handling
- Crops (and crop rotations)
- Field Fertilizer Practices
- Field Manure Practices
- Field Tillage Practices
- Field Soil Profiles
- Overall Field Management
- General RuFaS Simulation Settings

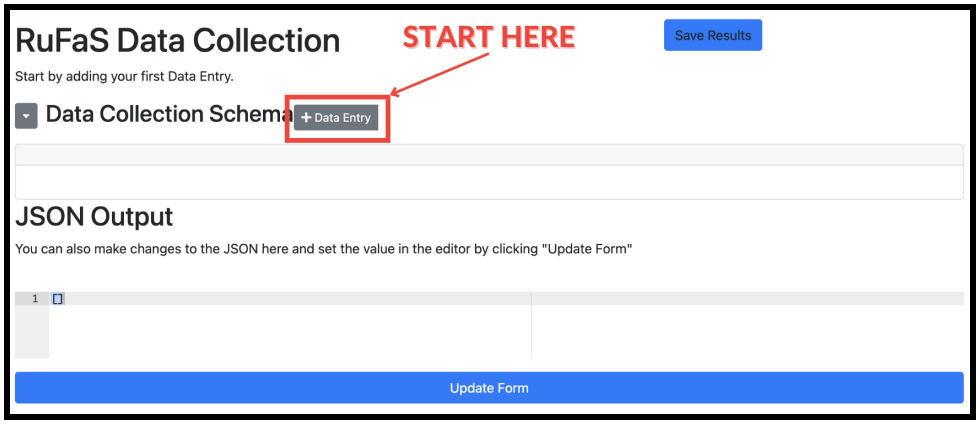
General Overview

- To start, double click on the index.html file which will open the app in your default web browser (usually it's one of Chrome, Firefox, Safari, Edge, or Internet Explorer). If the app is not behaving as described here, try running it with another browser.

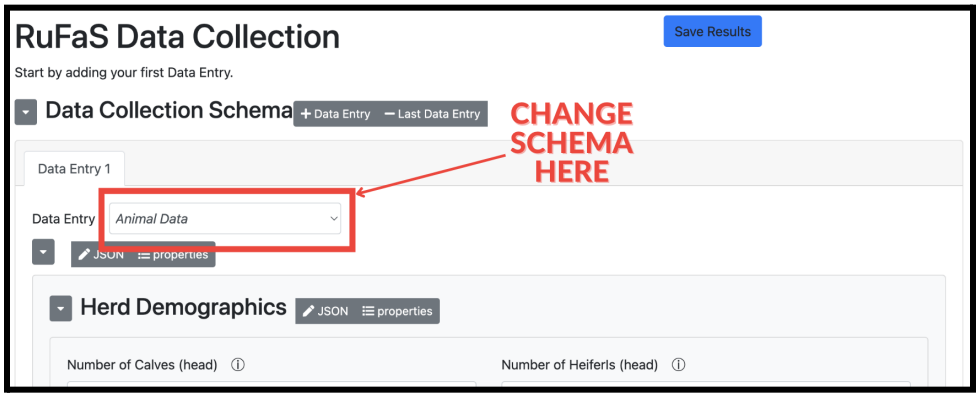


- Add a new Entry by clicking on the

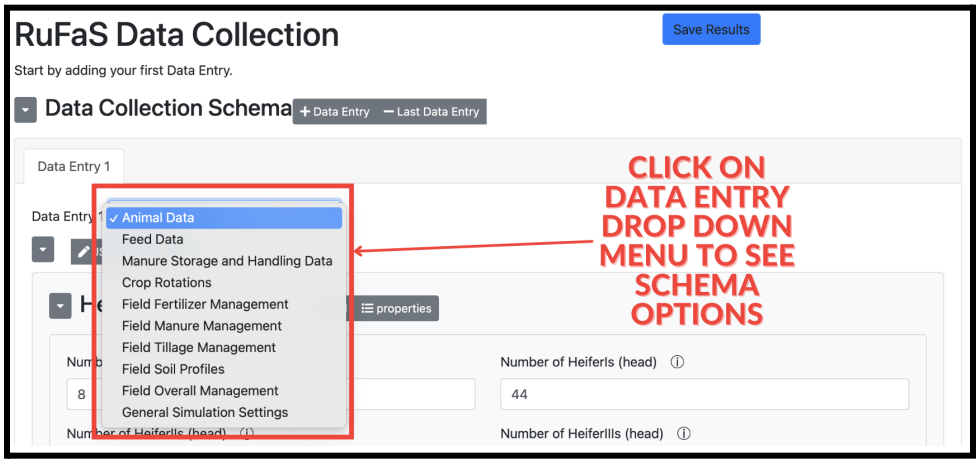
button



- The default schema for a Data Entry is *Animal Data.* Use the drop-down menu to change the schema:

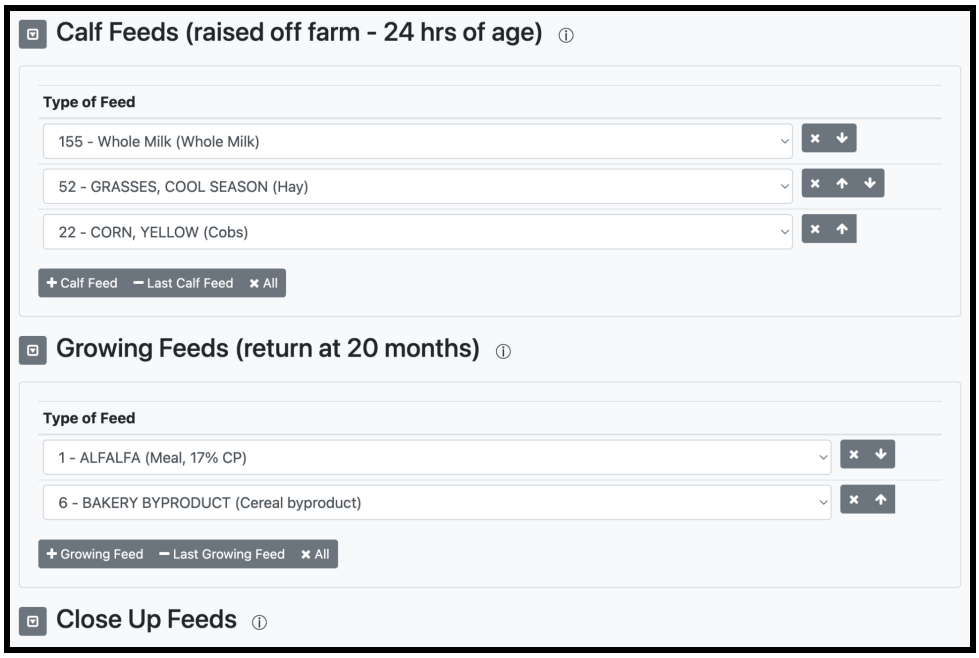


- All schemas are mandatory unless a specific entry does not apply to the farm. Each schema should be unique and should allow you to collect multiple types of the same data in that one schema. This means that there shouldn't need to be duplicate schema types (e.g. there shouldn't need to be 2 Field Soil Profile schemas because one Field Soil Profile schema can capture data on multiple soil profiles).

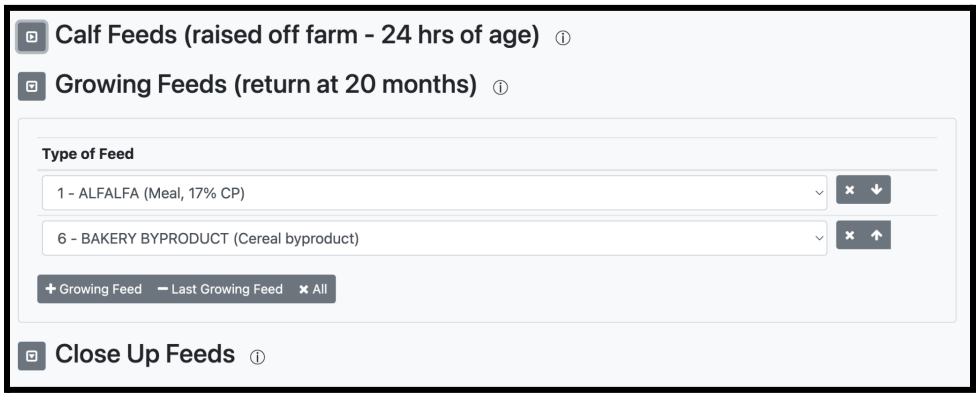


- After completing all the schemas, the user should click the “Save Results” button to save the results of their data collection to send to the RuFaS team (*more instructions on what to do with the file generated with your results at the end of this file*).
- Sections of a schema can be collapsed using the arrow buttons to reduce clutter and help keep data entry organized. They can also be expanded using the same button.

Calf Feeds section expanded:



Calf Feeds section collapsed (*Calf Feeds* data still present but hidden):



- Click the

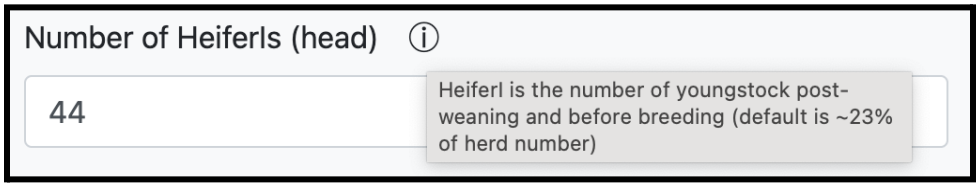
again to create the next schema (and the next after that, etc). It will save all previous data entered and it can be referenced anytime.

Filling Out the Data Entry Schemas

Many of the fields in the schemas are pre-filled with default values which is a typical value for that field - it may not be applicable to your particular farm! That data can be changed with your own data or left as is if you don't have a value for that particular field.

As you're going through the schemas you will see many of these symbols ⓘ by field input boxes - these contain further definition of what data we're looking for in that box, explanations of any acronyms used in that field name, or other contextual information. ***There is a list of all fields by schema in the appendix at the end of this user guide.***

You can see the information by hovering over the icon:



Animal Data

- Herd demographics (head counts of different animals classes)
- Breeding schedule and milk production
- Calf management programs and schedule
- Reproductive Program Management
- Bodyweight data
- Pregnancy check and loss data

Feeds Data

- Feeds used for different animal groups
- Feeds grown on farm
- Storage options used
- Purchased feeds

Manure Storage and Handling Data

- Manure handlers used along with water use and cleaning data
- Water use in milking, holding, and sprinkler systems
- Manure storage scenarios including:
    - Bedding type
    - Manure treatment process
    - Manure removal
    - List of pens using each particular manure storage scenario and what type/how many of each animal group is using that particular scenario

Fields Schemas

There are 6 Field schemas that get filled out for the farm:

- *Crop Specifications*
- *Field Fertilizer Management*
- *Field Manure Management (specifically for manure in field use)*
- *Field Tillage Management*
- *Field Soil Profiles*
- *Overall Field Management*

The 6th schema (*Overall Field Management)* is where practices in each of the first 5 categories are grouped together by which fields they're used on. For example:

- One field may use one soil profile, two fertilizer practices, one tillage practice, no manure practices, and uses a corn/alfalfa rotation.

- A second field may use the same soil profile as the first but uses one fertilizer practice, no manure or tillage practices and a corn/soybean rotation is used.

For this reason, **we suggest starting the Fields section by filling out the first 5 field schemas listed above** - to gather all the *Crop Specifications* used on the farm, all the different *Fertilizer*, *Manure*, and *Tillage Management* practices used, and however many different *Soil Profiles* there may be.

Then move to the *Overall Field Management* schema and organize fields by listing which of these practices are used on which fields. One set of practices may be used on a single field or a group of multiple fields. Multiple field uses can be grouped together in this way because they share the same set of practices and profiles.

Crop Specifications

- Species of crops grown
- Planting years each crop is grown
- Planting day
- Harvest years
- Harvest scheduling (“optimal” or “scheduled”)
- Harvest day (only necessary if Harvest Schedule is set to “scheduled”)

Field Fertilizer Management

- Fertilizer mix (NPK)
- Application schedule
- Mass of fertilizer used in an application
- Depth of the application
- Surface coverage

Field Manure Management

- Fertilizer mix (NPK)
- Application schedule
- Mass of manure used in an application
- Depth of the application
- Surface coverage

Field Tillage Management

- Tillage schedule
- Tillage method

Field Soil Profiles

- Soil geographic location
- Soil order classification
- Manning’s Roughness Coefficient
- Albedo reflectiveness
- Soil content (sand, silt, rock, clay)
- Depth
- Nitrogen mineralization

- Soil layer data (nice to have but not required):
    - Bottom depth
    - Bulk density
    - Field capacity
    - Layer content
    - Wilting and saturation points
    - Conductivity

Overall Field Management

This schema groups together practices from the previous 5 Field schemas:

- Field location
- Field area (covering multiple fields with the same crop, fertilizing, tillage, etc practices)
- Average slope % and slope length
- Initial residue thickness
- Practices listed in previous 5 schemas for each field/grouping of fields:
    - Crop(s) grown
    - Soil profile(s)
    - Fertilizer Specification(s)
    - Manure Specification(s)
    - Tillage Specification(s)

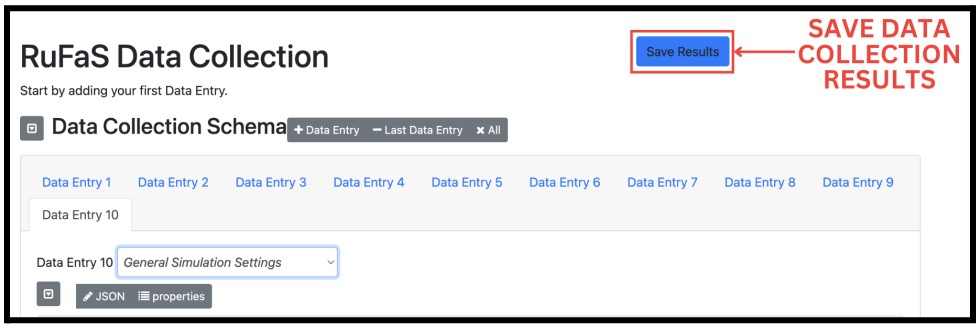
General Simulation Configuration

This schema contains settings used by the RuFaS model and only 2 pieces of data are collected from users:

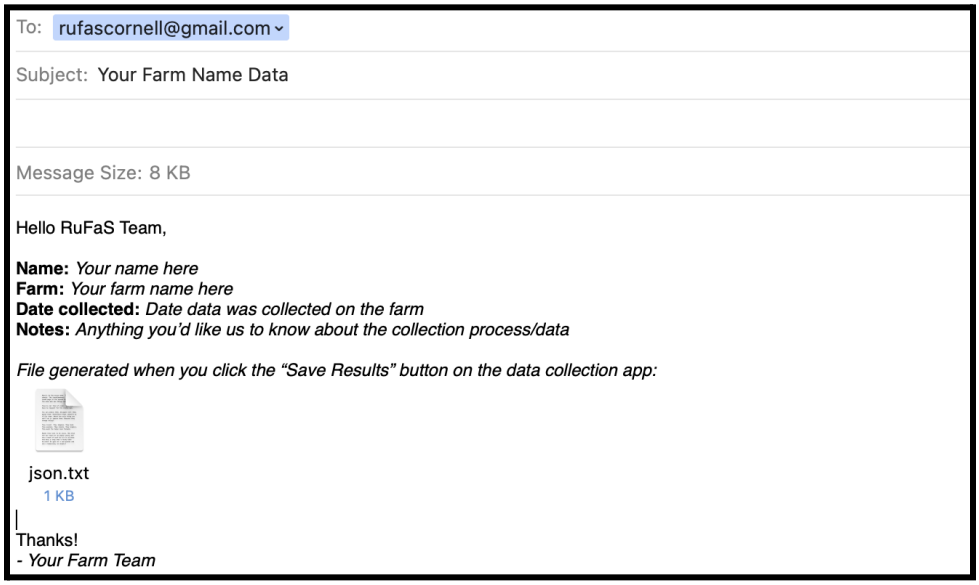
- Simulation Start Date
- Simulation End Date

Finishing up

The final step is to save the results:



Send an email to rufascornell@gmail.com **with the results file attached** and some basic information about the farm.



Template (for easy copy-paste into an email):

Hello RuFaS Team,

Name: *Your name here*
Farm: *Your farm name here*
Date collected: *Date data was collected on the farm*
Notes: *Anything you'd like us to know about the collection process/data*

File generated when you click the "Save Results" button on the data collection app:

Thanks!

- Your Farm Team

Appendix A: Input-Field Definitions

Animal Data

Herd Demographics

- **Number of Calves** - Average number of pre-weaned calves (head count)
- **Number of HeiferIs** - Number of youngstock post-weaning and before breeding (head count)
- **Number of HeiferIIs** - Number of heifers eligible for breeding and in early pregnancy (head count)
- **Number of HeiferIIIs** - Number of close-up heifers (head count)
- **Number of Cows** - Number of dry and lactating cows (all parities, head count)
- **Herd Number** - Number of lactating and milking cows on farm (head count)
- **Initialize Herd** - Signal to RuFaS to simulate the herd rather than using collected data - applicable only when cow body weight and milk production are exceedingly abnormal
- **Breed** - Predominant breed of the herd (Jersey/Holstein)

General Management

- **Breeding Start Day - Heifers** - Target start days born of reproduction programs for heifers (days)
- **Heifer Reproductive Program** - Reproductive program instituted with heifers (TAI, ED, or ED-TAI)
    - TAI - Timed Artificial Insemination
    - ED - Estrus Detection
    - ED-TAI - Combination of ED and TAI
- **Cow Reproductive Program** - Reproductive program instituted with cows (TAI, ED, or ED-TAI)
    - TAI - Timed Artificial Insemination
    - ED - Estrus Detection
    - ED-TAI - Combination of ED and TAI
- **Semen Type** - Type of semen that makes up a majority of the breedings on the farm (sexed or conventional)
- **Days in Pregnancy When Dry** - Average day in pregnancy when a cow is dried off (days)
- **Heifer Reproductive Cull Time** - Days old when a heifer would be culled for unsuccessful in breeding (days)
- **Do Not Breed Time** - Days in pregnancy when reproduction programs stop (days)
- **Cull Milk Production** - Minimum daily milk production for do-not breed cow to be culled (kg/d)

- **Number of Times Cows Milked Per Day** - Average or most common number of times the cows are milked (only whole numbers 1-3)

Farm Level Management

Calf Management

- **Male Calf Rate - Sexed Semen** - Rate of male calves when using sexed semen
- **Male Calf Rate - Conventional Semen** - Rate of male calves when using conventional semen
- **Female Calf Rate** - Rate female calves are kept and raised on-farm
- **Wean Day** - Days the calf is fully weaned from milk or milk-replacer (days)
- **Wean Length** - Days weaning (days)
- **Milk Type** - Type of milk used to feed calves (whole or replacer)

Repro Management

- **Estrus Detection Rate** - Percentage of in-heat animals that would be detected in the Estrus Detection programs
- **Estrus Service Rate** - Percentage of detected in heat animals that would be serviced in Estrus Detection programs
- **Heifer Timed Artificial Insemination (TAI) Protocol** - Breeding protocol for heifers (ED, TAI, or Synch-ED)
- **Heifer Conception Rate for md5CG2P Protocol** - Conception rate for md5CG2P protocol in heifer TAI (Timed Artificial Insemination) program
- **Heifer Conception Rate for md5CGP Protocol** - Conception rate for md5CGP protocol in heifer TAI (Timed Artificial Insemination) program
- **Heifer SynchED Protocol** - Either 2P or CP
    - 2P - Two injections of PGF - prostaglandin, watch for estrus after each injection with insemination upon heat detection
    - CP - 7 days CIDR implements, inject PGF upon removal, insemination upon heat detection
- **Cow Presynch Protocol** - Presynch protocol (1st breeding) for cow in TAI (Timed Artificial Insemination) program
    - Double OvSynch - GnRH-PGF-GnRH - TAI
    - Presynch - 2xPGF - TAI
    - G6G - PGF-GnRH - TAI
- **OvSynch56 Conception Rate** - Conception rate for OvSynch56 protocol (pre-breeding GnRH shot given 56 hrs post prostaglandin) in cow TAI (Timed Artificial Insemination) program
- **OvSynch48 Conception Rate -** Conception rate for OvSynch48 protocol (pre-breeding GnRH shot given 48 hrs post prostaglandin) in cow TAI (Timed Artificial Insemination) program
- **CoSynch72 Conception Rate** - Conception rate for CoSynch 72 protocol in cow TAI (Timed Artificial Insemination) program

- **CoSynch5d Conception Rate** - Conception rate for CoSynch 5d protocol in cow TAI (Timed Artificial Insemination) program
- **Cow Resynch Protocol** - Resynch protocol for cow that is diagnosed open at pregnancy in the TAI (Timed Artificial Insemination) program
    - TAI after PD
    - TAI before PD
    - PGFatPD"
- **Voluntary Waiting Period** - If TAI (Timed Artificial Insemination) is chosen for cow, it is days after parturition when the 1st hormonal injection is made in TAI programs. If 'ED-TAI' is chosen for cow, it is days after parturition when estrus detection followed by AI in ED-TAI programs
- **TAI Program Start Day** - Days after parturition when TAI (Timed Artificial Insemination) protocol starts in ED-TAI programs (days)
- **Conception Rate Decrease** - Decrease of the conception rate for later breeding
- **Average Gestation Length** - Average length in days of gestations (days)
- **Gestation Length Standard Deviation** - Standard deviation of gestation length (days)
- **Prefresh Day** - Length of period of the transition before calving (days)
- **Reproductive Indicator Calculation** - Measured in days (days)
- **Calving Interval** - For bodyweight adjustment and other calculations (days)

Bodyweight

- **Average Holstein Birth Weight** - Average birth weight for Holstein cows (kg/head)
- **Holstein Birth Weight Standard Deviation** - Standard deviation birth weight for Holstein cows
- **Average Jersey Birth Weight** - Average birth weight for Jersey cows (kg/head)
- **Jersey Birth Weight Standard Deviation** - Standard deviation birth weight for Jersey cows
- **Target Heifer Pregnancy Day** - Target pregnant age in days for heifers - for adjusting heifer body weight (days)
- **Average Mature Body Weight** - Average mature body weight of cows (kg/head)
- **Mature Body Weight Standard Deviation** - The standard deviation of mature body weight of cows (kg/head)

From Literature (Repro Values)

- **Pregnancy Check Day 1** - Days Since Last Heat of pregnancy check 1 (days)
- **Pregnancy Loss Rate 1** - % of cows that abort their pregnancy before the 1st pregnancy check
- **Pregnancy Check Day 2** - Days Since Last Heat of pregnancy check 2 (days)
- **Pregnancy Loss Rate 2** - % of cows that abort their pregnancy between the 1st and 2nd pregnancy check
- **Pregnancy Check Day 3** - DCC of pregnancy check 3 (days)
- **Pregnancy Loss Rate 3** - % of cows that abort their pregnancy between the 2nd and 3rd pregnancy check

Feeds Data

Feed Database Config

- **Feed Database/Feeds Table/Feed Quality Table/Nutrient Table** - settings needed for RuFaS Feed module setup and configuration (not changeable by user currently)

Animal Feeds

- **Types of Calf Feeds** - Feeds for calves (raised off farm - 24 hours of age)
- **Types of Growing Feeds** - Feeds for HeiferIs and HeiferIIs (return at 20 months)
- **Types of Close-Up Feeds** - Feeds for HeiferIIIs and Dry Cows.
- **Types of Lactating Cow Feeds** - Feeds for Lactating Cows

Farm Grown Feeds

- **Type of Feed** - Species and details of feed grown on farm
- **Storage Type** - Container in which the feed is stored
- **Additive** - Any substance added to the feed during storage that is beneficial to production
- **Inoculation** - Bacteria type used in feed inoculation
- **Ventilation** - Whether or not ventilation is used in the storage container

Purchased Feeds

- **Type of Feed** - Species and specs of purchased feed
- **Cost of Feed** - Cost of the purchased feed (US dollars/kg dry matter)

Manure Storage and Handling Data

Manure Handler Type

Flush System

- **Flush System Type** - Barn alleyways are cleaned by either fresh or recycled flush water from storage/lagoon
    - Alley flushing
    - Holding area flushing)
- **Water Type** - Type of water used in the flush system
    - Fresh
    - Recycled
- **Water Usage** - Amount of water use in the flush system (gal/animal/day)

- **Cleanings** - Number of cleanings per day

Scrape System

- **Scrape System Type** - Barn alleyways cleaned by automatic alley scrapers with minimal or no water added
    - Auto alley
    - Manual skid
- **Water Type** - Type of water used in the scrape system if applicable
    - Fresh
    - Recycled
- **Water Usage** - Amount of water use in the scrape system if applicable (gal/animal/day)
- **Cleanings** - Number of cleanings per day

Compost System

- **Bedding Type** - Material used for bedding pack
    - Sawdust
    - Dry wood shavings
    - Other
- **Other Compost Bedding Type** - Specific bedding type if selected “other” in previous column
- **Amount of Bedding Added** - Average amount of bedding added (kg/animal/day)
- **Frequency of Bedding Addition and Turning** - How often bedding is added or turned
    - Weekly
    - Biweekly
    - Other)
- **Other Bedding Frequency** - Specific bedding change frequency if selected “other” in previous column

Water Use

Milking Center Water Usage

- **Hose Pipe/Flush** - Total water usage during milking activity in the milking parlor (gal/animal/day)
- **Water Type** - Type of water used in the milking parlor system
    - Fresh
    - Recycled

Holding Center Water Usage

- **Hose Pipe/Flush** - Total water usage in the holding area before milking (gal/animal/day)
- **Water Type** - Type of water used in the milking parlor system
    - Fresh
    - Recycled

Sprinkler Water Usage

- **Sprinklers On Time** - 30 min cycle is assumed with an ON-time range of 1 to 5 min per cycle and an OFF time for the remaining time (minutes per cycle)
- **Sprinklers Off Time** - 30 min cycle is assumed with an ON-time range of 1 to 5 min per cycle and an OFF time for the remaining time (minutes per cycle)
- **Number of Timer Cycles** - 30 min cycle is assumed with an ON-time range of 1 to 5 min per cycle and an OFF time for the remaining time (cycles/day)
- **Water Flow Rate Per Cycle** - Water flow rate during the cycle (liters/min)
- **Sprinkler Set Point** - Temperature at which the sprinkler system is set to turn on and off within a cycle (degrees Fahrenheit)

Manure Storage Scenario Settings

Manure Scenario

- **Bedding Type** - Type of material used in the bedding
    - Sawdust
    - Sand
    - Manure-solids
    - Straw
- **Manure Handler** - Type of manure handler used
    - Manual scraping
    - Flush system
    - Automatic-alley scraping
- **Pre-Manure Separator** - Type of solid-liquid separator equipment to separate coarse fibrous solids/sand - prior to first treatment
    - Screw press
    - Sand lane
    - Rotary screen
    - None
    - Other
- **Other Separator Option** - If selected 'other' in previous column, write manure separator type here
- **Primary Manure Treatment** - First Manure Treatment
    - Slurry storage underfloor
    - Slurry storage outdoor

    - Composting
    - Anaerobic lagoon
    - Anaerobic digestion
    - Other
- **Other Treatment Option** -If selected 'other' in previous column, write manure treatment type here
- **Post-Manure Separator** - Type of solid-liquid separator equipment to separate digested/treated solids from the primary manure treatment - after primary treatment
    - Screw press
    - Sand lane
    - Rotary screen
    - None
    - Other
- **Other Manure Separator -** If selected 'other' in previous column, write manure separator type here
- **Final Manure Treatment** - Final Manure Treatment
    - Slurry storage underfloor
    - Slurry storage outdoor
    - Composting
    - Anaerobic lagoon
    - Anaerobic digestion
    - Other
- **Other Treatment Option** -If selected 'other' in previous column, write manure treatment type here
- **Manure Solids Handling** - How farm handles separator solids
    - Recycled as bedding
    - Land applied
    - Sold off-farm
    - Other
- **Other Manure Solids Handling** - If selected 'other' in first column, write manure solids handling option here
- **Manure Storage Removal Type** - Type of manure storage removal
    - Slurry pit storage under floor
    - Slurry pit storage outdoor
    - Effluent removal
    - Sludge solids removal)
- **Removal Period** - Length of period of removal - set value here and interval next (how long between manure storage removals)
- **Removal Period Interval** - Period length interval - tied to previous question
    - Days
    - Months
    - Years

Pens Using This Manure Scenario

- **Animal Grouping in this Pen** - Type of animals in this pen
    - CALF - Calves
    - GROWING - Heifer I & Heifer II
    - CLOSE_UP - Heifer III & Dry Cows
    - LAC_COW - Lactating Cows
- **Pen Size** - Number of animals in this pen (head count)

Crop Specifications

- **Species** - What type of crop
- **Planting Years** - Year(s) in which the planting of this crop occurs
- **Planting Days** - Julian day(s) of the year on which planting of this crop occurs
- **Harvest Years** - Year(s) in which the harvesting of this crop occurs
- **Harvest Scheduling** - Determines if a crop should be harvested on a fixed calendar-date schedule ("scheduled") or if harvest operations should be conducted to coincide with the highest yield ("optimal")
    - Scheduled
    - Optimal
- **Harvest Days** - Julian day(s) of the year on which harvesting of this crop occurs (not necessary to fill out if previous input “Harvest Scheduling” is set to “Optimal”).

Field Fertilizer Management

- **Fertilizer Mix** - Nitrogen-Phosphorus-Potassium nutrient mix of the fertilizer used
    - 0N_0P_60K
    - 6N_24P_24K
    - 5N_10P_30K
    - 5N_14P_42K
    - 82N_0P_0K
    - Custom_mix
- **Custom Mix Breakdown** - If selected 'custom_mix' in previous column, use format %N_%P_%K to specify custom nutrient mix for fertilizer

****Use only 1 of the following 2 application Input Options to describe when and how fertilizer is applied.****

Application Spec Input Option A (specifying fertilizer application years)

- **Years** - Years in which the application occurs (include start year)

OR

Application Spec Input Option B (specifying first year of fertilizer application followed by how many repeats of this application, followed by how many years to skip between repeats, followed by how many times the application cycle should be applied)

- **Start Year** - First application year
- **Repeat** - How many times should the application be repeated (after first) before skipping?
- **Skip** - How many years to skip before restarting the application pattern?
- **Cycles** - How many times should the entire application pattern repeat?

****The rest applies regardless of which Application Spec Input Option was used.****

- **Fertilizing Day** - Relative day on which fertilizer is applied
    - More than 7 days prior to planting
    - 1-7 days prior to planting
    - At planting
    - 1-7 days post planting
    - More than 7 days post planting
    - On specific date (please enter in next column)
- **Specific Start Day** - Julian day on which fertilizer is applied (only use if selected “on specific date” in previous column. Otherwise leave blank)
- **Fertilizer Mass** - Amount of this fertilizer applied at each application (kg/acre)
- **Fertilizer Depth** - Depth at which fertilizer is applied on field (cm)
- **Surface Coverage** - % of field cover by the application

Field Manure Management

- **Manure Mix** - Nitrogen-Phosphorus-Potassium nutrient mix of the manure used
    - 0N_0P_60K
    - 6N_24P_24K
    - 5N_10P_30K
    - 5N_14P_42K
    - 82N_0P_0K
    - Custom_mix
- **Custom Mix Breakdown** - If selected 'custom_mix' in previous column, use format %N_%P_%K to specify custom nutrient mix for manure

****Use only 1 of the following 2 Application Input Options to describe when and how manure is applied.****

Application Spec Input Option A (specifying manure application years)

- **Years** - Years in which the application occurs (include start year)

OR

Application Spec Input Option B (specifying first year of manure application followed by how many repeats of this application, followed by how many years to skip between repeats, followed by how many times the application cycle should be applied)

- **Start Year** - First application year
- **Repeat** - How many times should the application be repeated (after first) before skipping?
- **Skip** - How many years to skip before restarting the application pattern?
- **Cycles** - How many times should the entire application pattern repeat?

****The rest applies regardless of which Application Spec Input Option was used.****

- **Manure Applications** - Julian day(s) on which manure application is administered
- **Fertilizing Day** - Relative day on which manure is applied
    - More than 7 days prior to planting
    - 1-7 days prior to planting
    - At planting
    - 1-7 days post planting
    - More than 7 days post planting
    - On specific date (please enter in next column)
- **Specific Start Day** - Julian day on which manure is applied (only use if selected “on specific date” in previous column. Otherwise leave blank)
- **Manure Mass** - Amount of this manure applied at each application (kg/acre)
- **Manure Depth** - Depth at which manure is applied on field (cm)
- **Surface Coverage** - % of field cover by the application

Field Tillage Management

****Use only 1 of the following 2 Application Input Options to describe when and how tillage is applied.****

Application Spec Input Option A (specifying tillage application years)

- **Years** - Years in which the tillage occurs (include start year)

OR

Application Spec Input Option B (specifying first year of fertilizer application followed by how many repeats of this application, followed by how many years to skip between repeats, followed by how many times the application cycle should be applied)

- **Start Year** - First application year
- **Repeat** - How many times should the application be repeated (after first) before skipping?
- **Skip** - How many years to skip before restarting the application pattern?
- **Cycles** - How many times should the entire application pattern repeat?

****The rest applies regardless of which Tillage Spec Input Option was used.****

- **Start Tillage Use Day** - Timing to till, relative to planting event
    - More than 7 days prior to planting
    - 1-7 days prior to planting
    - At planting
    - 1-7 days post planting
    - More than 7 days post planting
    - On specific date (please enter in next column)
- **Specific Tillage Start Day** - Julian day on which tillage occurs (only use if selected “on specific date” in previous column. Otherwise leave blank)
- **Tillage Method** - Method by which the field was tilled

Field Soil Profiles

Whole Soil

- **Latitude & Longitude** - Geographic location of the soil in latitude and longitude (degrees)
- **Soil Order** - Soil classification based on US Dept of Agriculture soil order system
- **Manning’s Roughness Coefficient** - Unitless coefficient representing the friction factor of the conduit used to calculate water's rate of flow (cubic ft/second)
- **Albedo Reflectiveness** - Fraction of light that is reflected by the soil surface
- **Sand Content** - Average fraction of soil comprised of sand
- **Silt Content** - Average fraction of soil comprised of silt
- **Clay Content** - Average fraction of soil comprised of clay
- **Rock Content** - Average fraction of soil comprised of rocks
- **Depth** - Maximum depth of soil profile (i.e. maximum rooting depth)
- **Nitrogen Mineralization** - Rate at which nitrogen is mineralized to nitrate (N03) (annual %)

Soil Profile Layer

- **Bottom Depth** - Distance (in cm) from the soil surface to the bottom of this layer
- **Bulk Density** - Moist bulk density of the soil (g/cm^3)

- **Field Capacity** - Water content of the soil at field capacity (mm water/mm soil)
- **Wilting Point** - Water content of the soil at permanent wilting point (mm water/mm soil)
- **Saturation Point** - Water content of the soil when completely saturated (mm water/mm soil)
- **Sand Content** - Average fraction of layer comprised of sand
- **Silt Content** - Average fraction of layer comprised of silt
- **Clay Content** - Average fraction of layer comprised of clay
- **Rock Content** - Average fraction of layer comprised of rocks
- **Saturated Conductivity** - Saturated hydraulic conductivity (mm/ha)
- **Carbon Content** - Fraction of layer comprised of organic carbon
- **Nitrogen Content** - Fraction of layer comprised of active nitrogen
- **Labile P** - Fraction of layer comprised of labile phosphorus
- **Organic Matter** - Fraction of soil comprised of organic matter

Overall Field Management

General Field Data

- **Latitude & Longitude** - Geographic location of the field in latitude and longitude (degrees)
- **Total Field Area** - Land area of the field (acres)
- **Initial Residue** - Thickness of the residue present in the field at the start of the simulation (cm)
- **Average Slope of Field** - Average slope (%) of the field
- **Average Slope Length of Field** - Average length of the slope of the field (meters)

Crops Grown on Field

Crop Rotation

- **Crop Spec Species** - species grown
- **Crop Spec Number** - number of the crop species from the Crop Specifications Data Entry schema
- **Cover Crop Species** - species of cover crop (only fill out if “cover crop” is picked from first input field, otherwise leave blank)

Soil Specifications - Must have at least 1 entry but if there are multiple Soil Profiles in this field, they can all be added individually here

- **Soil Profile -** Number of the soil profile from the Field Soil Profiles Data Entry schema that is present in this field

Fertilizer Specifications - Not required if fertilizer isn't used on this field but if there are multiple Fertilizer Practices used in this field, they can all be added individually here.

- **Fertilizer Practice** - Number of the fertilizer practice from the Field Fertilizer Management Data Entry schema

Manure Specifications - Not required if manure fertilizer isn't used on this field but if there are multiple Manure Practices used in this field, they can all be added individually here.

- **Manure Practice** - Number of the manure practice from the Field Manure Management Data Entry schema

Tillage Specifications - Not required if tillage isn't used on this field but if there are multiple Tillage Practices used in this field, they can all be added individually here.

- **Tillage Practice** - Number of the tillage practice from the Field Tillage Management Data Entry schema

General Simulation Settings

Simulation Configuration

- **Weather Configuration File/Output Configuration File/CSV Directory Location/Graphics Directory Location/Set Seed/Seed** - settings needed for RuFaS setup and configuration (not changeable by user currently)
- **Simulation Start Date** - Date on which user wants the simulation to start
- **Simulation End Date** - Date on which user wants the simulation to end